

FEASIBILITY STUDY CAMPUS MARKETPLACE

AI-Powered College Second-Hand Marketplace

Project Overview: Campus Marketplace is a college-exclusive platform that enables students to buy and sell second-hand products such as books, calculators, electronics, cycles, furniture and stationery. The system provides product listing, search, buyer-seller interaction and administration features, enhanced with AI-based price recommendation, semantic search, suspicious-listing detection, and a commission system for the platform administrator.

A. Technical Feasibility

The project is technically feasible using mature and accessible technologies. The frontend will use **React.js** and Tailwind CSS, while the backend will use **Node.js** and **Express.js**. **MongoDB** will manage marketplace data and Cloudinary can be used for product images. The AI layer will use **Python**, **Scikit-learn**, **XGBoost**, **FastAPI** and Sentence Transformers. AI price prediction will estimate suitable resale prices for sellers and evaluate price fairness for buyers. Semantic search will retrieve relevant products even when search terms do not exactly match listings. Suspicious-listing detection will flag unusual pricing and activity for administrator review. Therefore, the required technology and technical skills are available.

B. Managerial Feasibility

The project is manageable for a four-member student team. Responsibilities can be divided into frontend development, back-end/API development, database and deployment, and AI/ML development. Git and GitHub will be used for collaboration and version control. Development will follow incremental stages, beginning with the core marketplace and subsequently integrating transactions, commission, and AI features. This division reduces workload and allows parallel development.

C. Economic Feasibility

The project has low development costs because most required technologies are open-source or provide free development tiers. React, Node.js, Python, Scikit-learn, FastAPI, GitHub and MongoDB can be used with minimal initial expenditure. The platform can also generate future economic benefits through an administrator commission on completed transactions. For example, a 5% commission on a transaction of INR 1,000 would generate INR 50 for the platform. Thus, the expected benefits justify the development effort.

D. Financial Feasibility

The academic prototype does not require significant external funding. Development can be performed using existing computers and free or student-level cloud services. Payment functionality can initially be implemented using a test/sandbox environment, avoiding real financial transactions. Production deployment would involve additional costs for hosting, payment processing, security, maintenance and compliance. Therefore, the project is financially feasible at the prototype level.

E. Cultural Feasibility

The platform is compatible with the existing culture of college students. Students already buy and sell used products through WhatsApp groups, social media, personal contacts and informal campus communities. Campus Marketplace simply organizes this existing behavior into a centralized platform with structured listings, searchable products and transparent pricing. Hence, users would not need to adopt an entirely new form of buying and selling.

F. Social Feasibility

The project can provide positive social benefits by making affordable second-hand products easier to access and allowing students to recover value from unused items. It encourages product reuse and can reduce unnecessary waste. AI-based price recommendations can improve price transparency, while suspicious-listing detection and administrative moderation can improve trust between buyers and sellers. The platform therefore has a positive potential impact on the campus community.

G. Safety Feasibility

The platform can incorporate authentication, role-based authorization, secure password storage and protected APIs to improve data security. The commission-based transaction workflow can provide additional financial protection by recording payments and transaction status. Suspicious listings will be flagged for human administrator review rather than automatically banning users. Users should also be encouraged to exchange products in safe public campus locations. Physical safety cannot be completely controlled by the software, but major digital and transactional risks can be reduced.

H. Political Feasibility

The project is politically feasible as an academic prototype using test-mode payments. A production deployment involving real payments would require appropriate business registration, payment-provider approval, KYC, taxation, data protection and institutional policies. These requirements do not prevent development of the proposed academic prototype but would need to be addressed before commercial deployment.

Go / No-Go Decision

GO

The feasibility analysis indicates that Campus Marketplace is technically, managerially, economically, financially, culturally, socially, safely, and politically feasible as an academic prototype. The project should be developed incrementally, starting with the core marketplace and followed by the commission system and AI capabilities.